SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Dinesh.V (192524309) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Dinesh.V
Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

The message given is:

MEET AT NOON

A fixed shift of 4 positions is applied

This is the caesar cipher, where every alphabetic character is shifted by the same no. of positions.

(a) Encryption using shift = 4

i.e., key = 4

A	B	C	D	E	F	G	H	I	J
0	1	2	3	4	5	6	7	8	9
K	L	M	N	O	P	Q	R	S	T
10	11	12	13	14	15	16	17	18	19
U	V	W	X	Y	Z				
20	21	22	23	24	25				

Plaintext :	M	E	E	T	A	T	N	O	O	N
value :	12	4	4	19	0	19	13	14	14	13
key = 4 +										
ciphertext value :	16	8	8	23	4	23	17	18	18	17

ciphertext : Q I I X E X R S S R

spaces remain unchanged.

∴

Plaintext : MEET AT NOON

ciphertext : QIIX EX RSSR

(b) Decryption:

An authorised member knows that the encryption uses a shift of 4.

∴ the receiver performs the reverse operation.

for example:

Φ → M

I → E

X → T

E → A

R → N

S → O

Thus :

ΦIX EX RISSR

↓ shift −4

MEET AT NOON ← Encrypted Message.

formula used in caeser cipher

Encryption : $C = (P+k) \bmod 26$

Decryption : $P = (C-k) \bmod 26$.

The authorised receiver can recover the original message by applying reverse shift of four positions.

The company uses keyword-based encryption technique.

Given:

Plaintext : HELLO

keyword : KEY

This is the vigenere cipher, which uses a repeating keyword to determine the shift applied to each plaintext character.

(a) Step-by-step encryption

First, repeat the keyword until it matches the length of the plaintext.

Plaintext : H E L L O

keyword : K E Y K E

Plaintext	H	E	L	L	O
Plaintext value	7	4	11	11	14
Key	k	E	Y	k	E
Key value	10	4	24	10	4
Ciphertext value	17	8	9	21	18
Cipher text.	R	I	J	V	S

⇒ Plaintext after encryption ⇒ RIJVS

(b) Effect of the repeating keyword

The keyword KEY is repeated to match the message length:

HELLO
KEYKE

Each keyword character determines the shift for the corresponding plaintext character.

Therefore:

plaintext	H	E	L	L	O
key	K	E	Y	K	E
ciphertext	R	I	J	V	S

Unlike caecer cipher uses same shift for every character, vigenere cipher uses different shifts based on the keyword. vigenere cipher belongs to poly alphabetic substitution.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

85

85/100

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